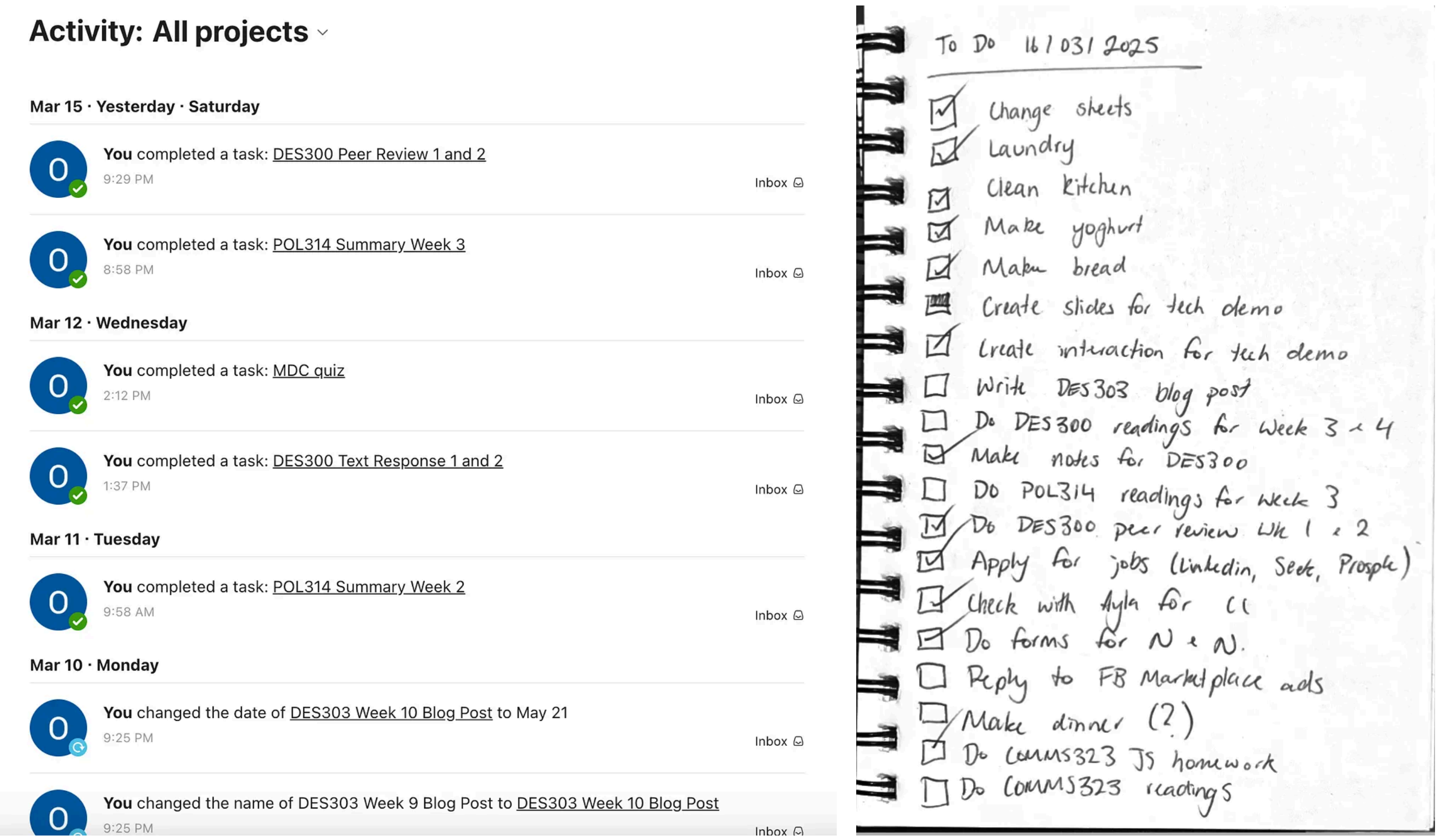
DES303 Week 2: Tech Demo Prep & Experiment Ideation

12 min read · Mar 17, 2025

Livi Lee Follow

Listen Share

I needed to hit the ground running this week. With our tech demos just over a week away, experiment ideation to do, and blog posts to write — plus a host of other readings and assignments for my other classes — it was essential that I approached the week with a solid plan.

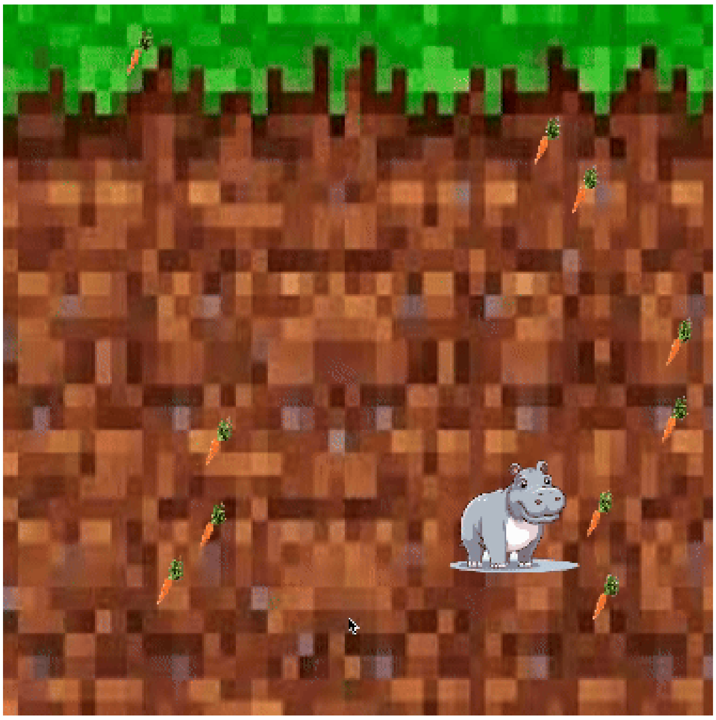


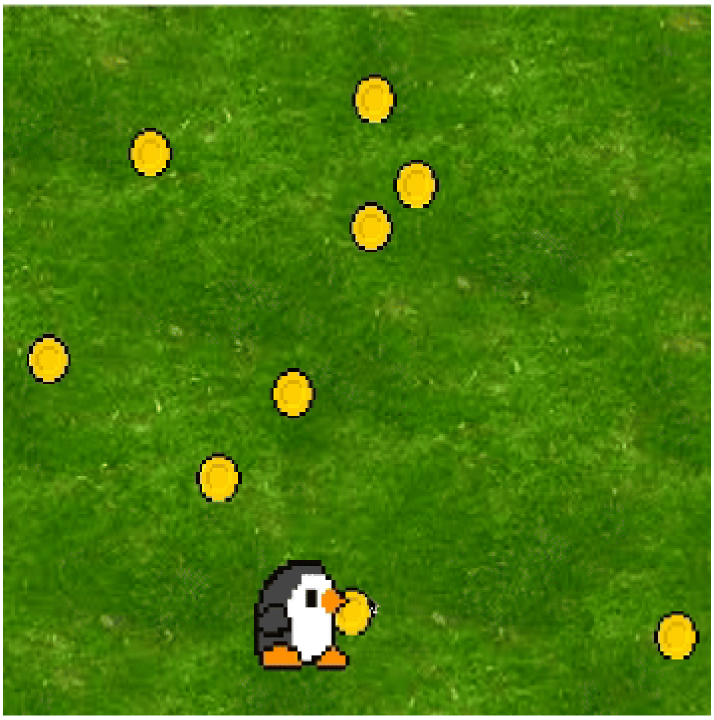
A busy week where keeping track of all my tasks was essential. My planning methods on ToDoist and on paper are working well for me so far.

Part I: Description

Tech Demo

Most of my work this week focused on the upcoming tech demo. I began by completing the interactive game that I'll use to demonstrate my examples during the tech demo. I also captured screen recordings of me playing the game and adapting the code, which I'll include in the Canvas version of my slides.





The same game with the asset links changed (ie. same game mechanics, different visuals). Showing my peers how simple it is to swap out visual assets will be a key part of my demo. However, this was just a testing stage and these visuals won't be used in the final product.

I've also created the bulk of my slide deck for the tech demo — the only section left is the technical explanation. I want to talk to my group members before completing this task, as I'm not sure their level of coding experience. It's important that my tech demo isn't too easy or too difficult, as I want it to be engaging and educational.

I ran into some challenges with getting the collision detection function in the code to work (this code detects when the character has 'collected' the coins or other items.) However, referring to the Processing documentation and a previous assignment I had done in Processing for DES240 helped me to find a solution.

Tech Demo

```
void checkPosition(){

  if (x > xPosition-80 && x < xPosition+80) {
       if (y > yPosition-80 && y < yPosition+80) {
        w = 0;
        caughtCheck = caught;
      caught = true;

      if (caught == true && caughtCheck == false){
    Counter += 1;
    caughtCheck = caught;
  }
  }
  }

  if (Counter == numItems){
    text(congratsText, 60, 120);
     }

}
}
```

This section of code was more complicated than I remembered! It detected whether the character had collected an item, counted it in the score, and checked whether all the items had been collected.

I also hit a bit of a snag when I realised that my group members would need to have Processing IDE downloaded on their computers to participate in the interactive part of the demo. While I could ask them to download it beforehand, this feels risky, particularly if there are any members of my group who don't show up, or someone extra who joins us on the day. I definitely don't have time during my 15-minute demo to wait for my peers to install a new application.

Originally, I wanted to send my code to each of my group members individually so that they could play around with it themselves. However, with this new realisation, I have three options: either do the demo entirely through prerecorded videos, do the demo the interactive parts myself with everyone watching, or have them do it on my computer.

At the moment, I think having my peers do the interactive part of the demo together on my computer is the best option. I think it's important that *they* get to try out being creative with the code, particularly as one of the key messages of my demo is that Processing is a really easy way to rapidly prototype scalable projects. While this will mean I don't have my digital notes accessible for the interactive part of the demo, I think this is a worthwhile trade-off.

[Note: as I got closer to the demo, I realised that I could send out an announcement to my peers asking them to download the software and my interactive code in advance. This should allow us to have a more streamlined demo experience, with user interaction! I've also prepared videos in case any of my peers don't see the announcement.]

Experiment Ideation

This week, I focused on building up my bank of experiment ideas. While I had done some mind-mapping and ideation, I was still feeling like I hadn't really gotten to the 'aha' stage of ideation yet. I was only scratching the surface! So, this week, I challenged myself to come up with 100 experiment ideas. Later, I'll refine this list and come up with an experiment plan; but this week, it's all about getting as many ideas down as possible.

I had some rules for the ideas that I wrote down, though:

- They had to have a specific output (ie. poster, UI screen, animation.) I knew that if I proposed a problem without a particular output, I would get stuck thinking about all the ways I could solve that problem.
- I had to *want* to do the experiment. It's easy to plug a prompt into ChatGPT and get it to spit out a list of experiment ideas — unfortunately, when I tried this, I wasn't interested in actually making any of the things it suggested. While I did use AI to inspire some of my ideas, I curated the lists to ensure that everything I was writing down was actually interesting and something I wanted to do. (I've noted where ChatGPT was used to inspire the ideas.)
- The experiment had to be feasible. I didn't have to know *exactly* how to do it, but it should be small enough to be achievable in a few days. If I didn't know how to do something, I should have a good action plan on how to tackle it (ie. YouTube tutorials.)

With all those rules, here is the list of experiments that I came up with over the last few days:

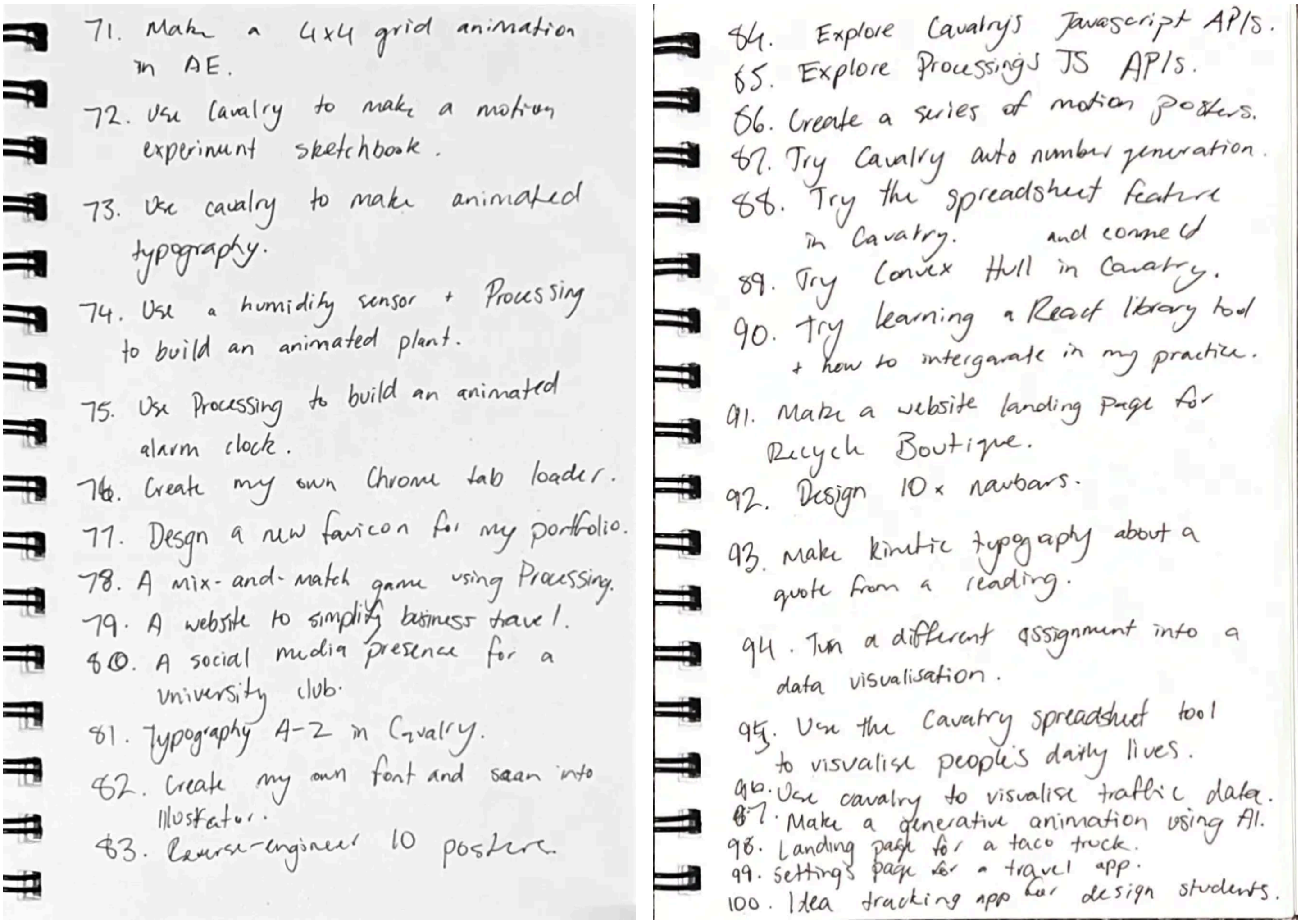
Can I think of 100 Design Experiment Ideas... that I would want to try?

I am [out-put focused].

1. A poster using mixed media advertising a news article.
2. An interactive poster (physical) about social media usage.
3. An interactive poster (digital) about data soveriegnty.
4. A set of playing cards with facts about Aucklanders on them.
5. A 2-D animated poster with 4 characters.
6. A ~~title~~ logo animation for an NZ brand.
7. UX design - an app for students to track their weekly readings.
8. An onboarding sequence for B.com
9. A pricing chart for ~~newspaper~~ app pottery classes.
10. 4 animated loading icons.

CHATGPT

11. Climate change dashboard.
12. Museum AR experience (?)
13. A new font in my style.
14. A graphic about AI energy use / water consumption.
15. A poster with every outfit I wear in a week.
16. An app - more visible screen time tracker.
17. An interaction that captures a physical input in an animation.
18. Personal expense UI for students.
19. Spotify Wrapped 2024 (but better)
20. Character walk cycle.
21. Fun buttons!
22. A logo for my portfolio.
23. Design + print a tote bag.
24. Papercut animation.
25. Figma × 4 options for ecommerce shopping page w pros + cons.
26. Use Cavalry to make a 10-sec animation about writing flow.
27. Design a lamp for my bedside table.
28. Design a poster to help people understand big data.
29. A new brand identity for (??)
30. An app for tracking pottery.
31. A poster using only two colours.
32. A hand-drawn animation about an orange.
33. A hand-drawn animation about air pollution.
34. An interactive map of liquor stores in Auckland.
35. A poster about jail in Auckland.
36. An infographic about climate change violence.
37. A poster using the graphics from the BLM movement in 2020.
38. A youtube introduction for a history channel.
39. Packaging design for a vegan protein drink.
40. Infographic about maternal death in NZ.
41. Poster using mixed media about Severance.
42. A poster that combines AI generated art.
43. A zine about being sick.
44. A poster inspiring me.
45. A pottery workshop where everyone's hands make their cup.
46. Animation showing shrinking news.
47. Buttons with micro-interactions.
48. Unique 404 page.
49. A new desktop wallpaper for me.
50. Generative collages
51. An animation using the copy function.
52. Combination of these two?
53. Visualisation of democracies around the world
54. Animation of the history of pots.
55. Guide to destroying a website.
56. A new game for the NYT.
57. An app showing grid layouts
58. Turn an ~~2D~~ image 3D in Processing.
59. Design a book cover.
60. Design a pinwheel in Spline.
61. Make light glow effects in PS.
62. Train trip booking website.
63. Reactive spheres in Spline.
64. 3D print topographical map
65. CNC topographical map
66. Interactions in Spline
67. Characters in Spline
68. Code an AI chatbot in Python
69. Use Procreate Dreams to make a uni-themed video.
70. Make a looping animation about COVID-19 in processing.



71. Make a 4x4 grid animation in AE.
72. Use Cavalry to make a motion experiment sketchbook.
73. Use cavalry to make animated typography.
74. Use a humidity sensor + Processing to build an animated plant.
75. Use Processing to build an animated alarm clock.
76. Create my own Chrome tab loader.
77. Design a new favicon for my portfolio.
78. A mix-and-match game using Processing.
79. A website to simplify business travel.
80. A social media presence for a university club.
81. Typography A-Z in Cavalry.
82. Create my own font and scan into Illustrator.
83. Reverse-engineer 10 posters.
84. Explore Cavalry's Javascript APIs.
85. Explore Processing's JS APIs.
86. Create a series of motion posters.
87. Try Cavalry auto number generation.
88. Try the spreadsheet feature in Cavalry. and connect
89. Try Convex Hull in Cavalry.
90. Try learning a React library tool + how to intergarate in my practice.
91. Make a website landing page for Recycle Boutique.
92. Design 10x navbars.
93. Make kinetic typography about a quote from a reading.
94. Turn a different assignment into a data visualisation.
95. Use the Cavalry spreadsheet tool to visualise people's daily lives.
96. Use cavalry to visualise traffic data.
97. Make a generative animation using AI.
98. Landing page for a taco truck.
99. Settings page for a travel app.
100. Idea tracking app for design students.

I definitely noticed some patterns emerging in this ideation exercise, particularly around what technologies I'm interested in using. I'll organise this list, analyse, and develop a full experiment plan in next week's blog post.

Part II: Feeling

Tech Demo

I'm feeling prepared for the tech demo on Thursday. Luckily, public speaking has never really bothered me, so I'm often less stressed out about presentations than some of my peers. I'm also looking forward to seeing what technologies my other group members are going to be presenting. Hopefully, their tech demos will give me some more ideas for future experimentation.

My biggest concern is my level of expertise in Processing itself. While I can usually figure out how to do what I want, I'm by no means an expert — I only started using it myself last year! Part of the reason I wanted to use Processing for my tech demo was to remind myself of the basics by preparing to teach about it. So, I'm hoping that there are no major technical errors, or tricky questions.

```
//***************************************************************************CHANGE ME!!**********************************************************

//Change these links to show different backgrounds, characters, and items to collect.
//I've preloaded some for you. Check out the file for all the options, or try changing the numbers in the file name to 2 or 3.
//For example: background-2.png

background = loadImage("background-3.png");
character = loadImage("character-3.png");
item = loadImage("item-3.png");

//***************************************************************************************************************************************************//
```

```
//****************************************************************************CHANGE ME

//Change this number to increase or decrease the number of items to collect.
int numItems = 10;

//Change this text to match the item that your character collects.
String itemTypeText = "rocket ships";

//*************************************************************************************
```

Experiment Ideation

I felt quite overwhelmed during the ideation phase of this task. There are so many things that I could make, and so little time to get them all done! While I'm grateful for this time to prepare before we start building out capstone projects, I can't help but kick myself for having not built some of these skills over the last few years.

Doing the 100 ideas helped me to feel less panicky. Once I saw all the ideas I had laid out, it was much easier to sort through them and choose the ones that I thought would be most beneficial. I could also make more of a plan about how each experiment would lead into another, rather than not knowing what I was going to be working on next.

Once I had got all of my ideas out on paper, I also felt quite excited to get started. After all, one of the criteria for these ideas was that I had to *want* to do them!

Part III: Evaluation

Tech Demo

Most of my plan for my tech demo feels solid; however, I still haven't completed the technical slides, where I do the bulk of the teaching! This part is really important to make sure that my peers understand the technology I'm demonstrating. I definitely need to spend some time on this over the next few days. I also need to try some practice run-throughs to see how long the presentation will take in total.

Apart from this, the process of preparing for the tech demo hasn't been too stressful — in fact, I've quite enjoyed revisiting Processing and thinking about how best to teach it to others.

Experiment Ideation

My ideation process, however, was a little bumpy. I started to overthink things in my head — trying to balance creating experiments, adding projects to my portfolio, testing new mediums, learning new tools and techniques, impressing future employers, and getting ready for Capstone — it was a lot to juggle! I had to keep reminding myself that my only goal here to to explore creative possibilities. I didn't have to do everything in this one project — in fact, it would probably turn out better if I kept focused on my goal.

I also ran out of ideas when doing my 100 ideas challenge at precisely Idea 52. Full disclosure — I thought about ending the challenge there and pretending I was only aiming for 50!

Unfortunately, the first-year class I'm TAing for is currently working on their 300 sketch assignment. After spending all morning telling students that they couldn't give up even when 300 felt like an impossible number, I would've felt like a hypocrite if I had given up halfway. So, I sat back down and came up with 48 more ideas.

Part IV: Analysis

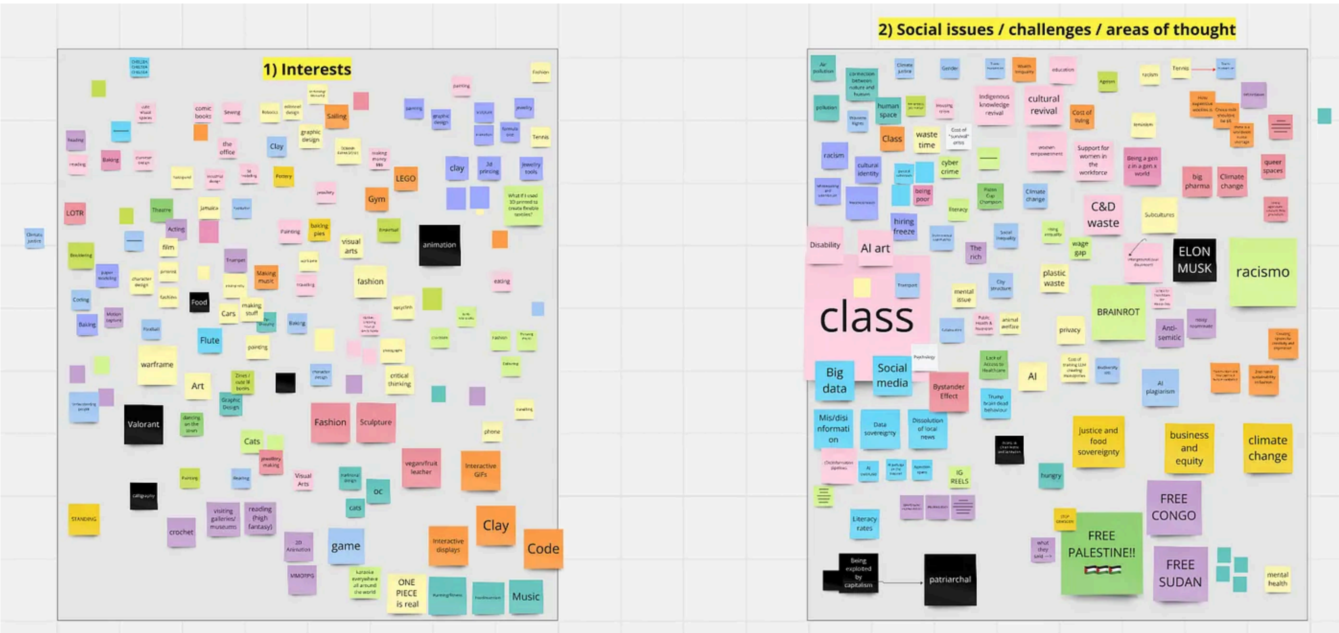
Tech Demo

I'm not sure why I'm procrastinating making the technical explanation slides. It could be because these are the most 'boring' bits — explaining variables and loops isn't the most thrilling topic, particularly if my peers have learned it in coding classes before. I don't think it'll take very long to make these slides as they are pretty straightforward — it's just actually sitting down to make them that's the challenging part!

I've noticed this tendency in my working pattern before — I get really excited about a project, work lots, and make lots of progress while far away from the deadline. I push the boring or difficult bits to the end of the project, and procrastinate them because 'the deadline is so far away!' I also tend to do the 'light-thinking' tasks first, which I can do while multitasking (ie. half-chatting with friends, or with a TV show on in the background.) Then, a few days before, I'm left scrambling to do the boring, more difficult tasks (which often end up being more time consuming than expected!)

Experiment Ideation

Why did I get so stuck with coming up with ideas? I had lots of sources of inspiration — I looked at Pinterest, asked ChatGPT, went back through the ideation session that the class did with Leo — but it was still a slow and painful process getting out 100 ideas.



Revisiting the Miro board we made in DES303, I found that the first two boards were most helpful for my ideation process. They helped me to break the pattern of the interests and issues I'm always thinking about, and see what else my peers are interested in.

I think using a laptop to help generate ideas may have stifled my creativity somewhat. I've noticed that it's much easier for me to come up with original ideas when in conversation with someone else, or when I'm outside, away from other distractions. However, because I was looking at inspiration on the Internet, it was easy to get distracted or become demotivated from looking at how talented other designers were (and how inadequate I felt!)

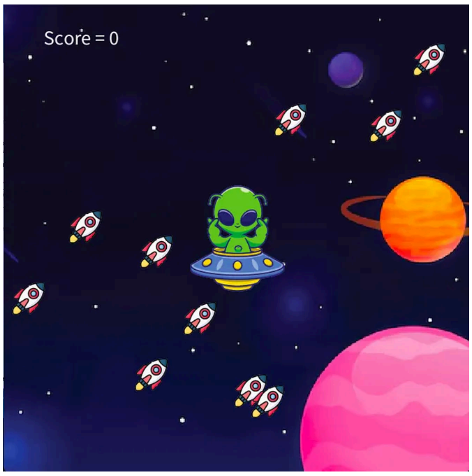
To my surprise, I also found that talking to my students in DES100 to be a major source of inspiration. While I was giving them advice on how to come up with ideas for their sketches, I could apply that advice to my own ideation process. For instance, I instructed one group of students to draw everything they were currently touching (what projects could I do about the items *I'm* currently touching?). I told another group to draw every logo they saw for an afternoon (could I redesign every logo *I* saw for the afternoon?) The process of taking these unfiltered ideas without judgement, and trying to create an experiment idea that I wanted to do with them, gave me a lot of inspiration.

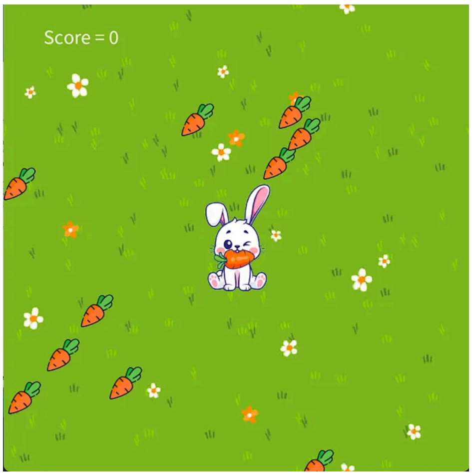
Part V: Conclusion

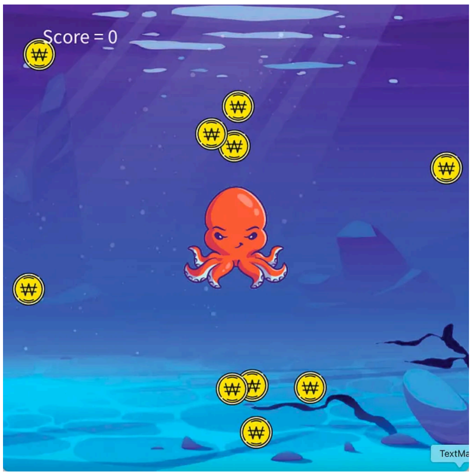
Tech Demo

Overall, planning the tech demo has gone relatively smoothly for me. As I write this, I've completed the interactive code and most of the presentation. I found that the time management processes (ie. ToDoist) that I set up last week are working really well for me to keep on top of all my classes and club activities. I've also learned that I've really enjoyed exploring Processing as a tool, and it could be a good medium for me to take further into Capstone.

However, writing this reflection has made me think about why I haven't completely finished the presentation, and why I've been putting off making the technical slides. I've realised that I have a tendency to push the most boring or difficult parts of a project to the end, disrupting the design process and ultimately making the project more difficult to complete. In the future, I think it would be beneficial for me to force myself to do my least favourite parts of a project first — the boring, technical, 'deep-thinking' tasks — and leave the more enjoyable creative and making tasks until I've completed these. I hope this will make it easier for me to get to the finish line on these projects.







Finding these graphic assets was fun, and coding in the different aesthetics was also really enjoyable! This is one of the first things I did for this project. However, in the future, I should keep fun tasks like this until the end of the project and prioritise the more difficult, boring, and time-consuming tasks.

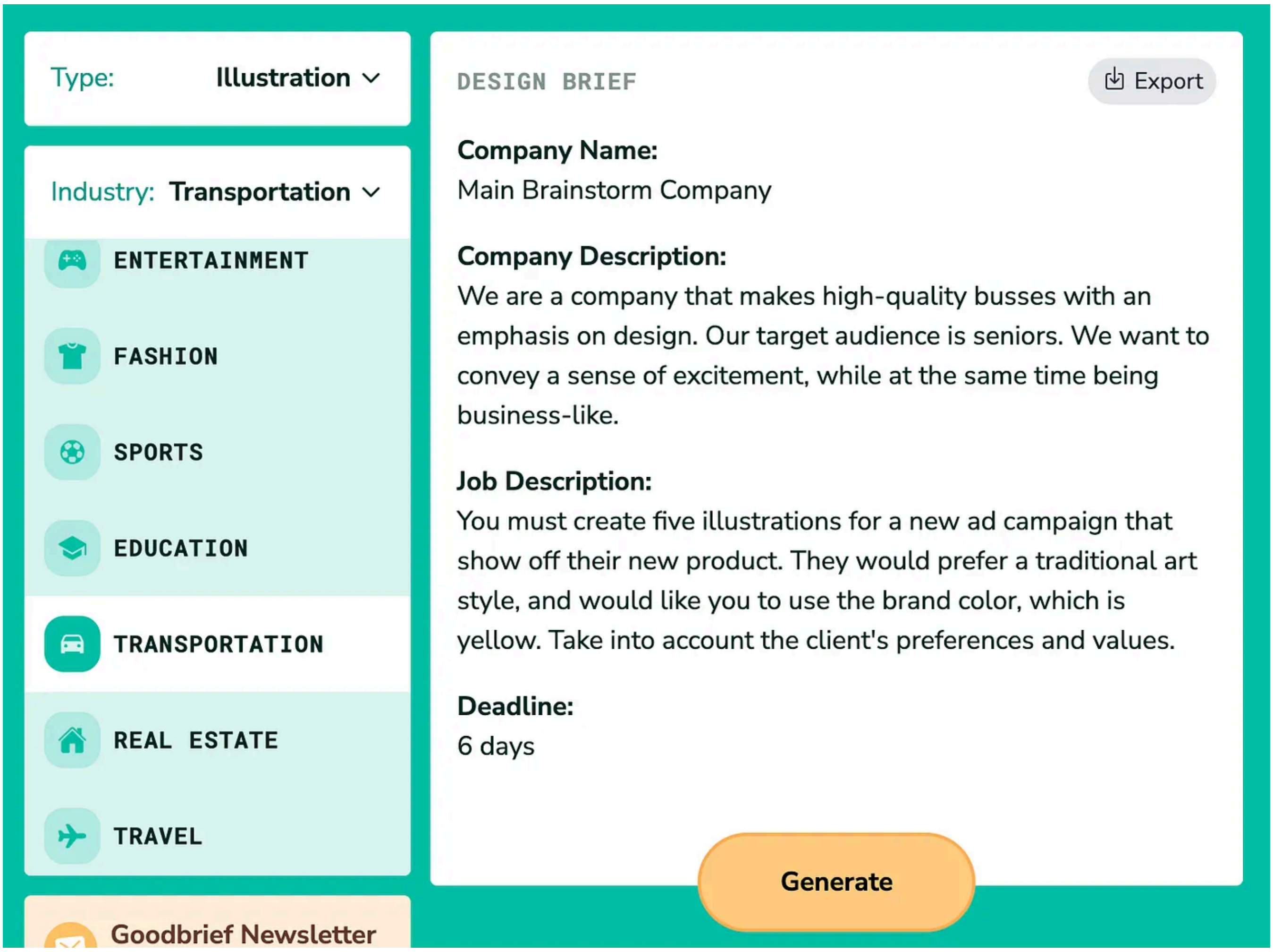
Experiment Ideation

Overall, I was satisfied by my ideation process. While I did get stuck when trying to come up with 100 ideas, once these had been completed it was reassuring to see all these different ideas to choose from laid out on paper. It helped to reduce my sense of overwhelm and clarify my plan for moving forward.

Some key insights that I found out about my ideation process this week:

- I find my best inspiration from conversations and being outdoors. I can easily get stuck when I'm only looking for inspiration online. This is important to keep in mind as I progress through the semester.
- Setting an ambitious goal of 100 ideas (and pushing through to complete it) gave me a sense of satisfaction and also improved the quality of my overall design process. I should continue to set ambitious goals and push myself to complete them throughout the semester, even when not explicitly required by the assignment.
- I felt most inspired when I focused on the final output I wanted to create (ie. a poster) instead of a undefined challenge. This will allow me to work faster and improve my skills more efficiently. I will save the 'undefined challenge' for my DES300 proposal work.
- I didn't find that AI generated responses to ideation prompts, or the online 'brief generator' tools, were very helpful at all. While they are good tools for mixing

and mashing ideas, I didn't *want* to make any of the things they suggested.



This randomly generated brief from https://goodbrief.io/ wasn't inspiring at all. The key words were replaced on each generation, MadLibs-style, but the final brief didn't make any sense.

Part VI: Action Plan

Tech Demo

To ensure that I complete the slides for my tech demo on time, I will do the following steps:

- After I finish writing this reflection, I will immediately make a draft of the technical slides for the tech demo. Calling these slides a 'draft' takes the pressure off of making them perfect, so I can get all my thoughts down before refining the slides.
- On Wednesday morning, when I have no classes, I will do four practice run-throughs of my tech demo. Three will be by myself (timed), and one will be to

my partner to see how long it takes someone who hasn't seen the code to do the tasks. I will adjust the presentation as needed based on this run-through.

For future projects, I will ensure that I do the most difficult and time-consuming tasks at the beginning of the project, rather than pushing it to right before the deadline.

Experiment Ideation

Over the next week, I will revisit the 100 ideas I have come up with, and discuss their potential with my friends and peers. I'll then choose a top 10 that I'm interested in making (saving the other ideas for later, just in case.) I will then categorise them into design practices (eg. graphic, animation, UX). I will make a calendar for the rest of the semester and plan out when I'm going to complete each experiment. These should feed into each other in terms of complexity — ie. completing the easier projects first to build skills before moving on to more difficult challenges.

To maintain my motivation, I will continue to set myself ambitious goals throughout the semester. For instance, for each project, I should create at least 5 possible concepts or iterations before deciding on a final direction.

In the future, at the beginning of each project, I will also put aside some time (approx. 1 hr) to sit and ideate with no screens, only pen and paper. Ideally (weather-dependent), this will be outdoors. This will help me to have a more focused ideation process.

Tech Demo Experimentation Design Processing

Follow

Written by Livi Lee

0 followers · 2 following

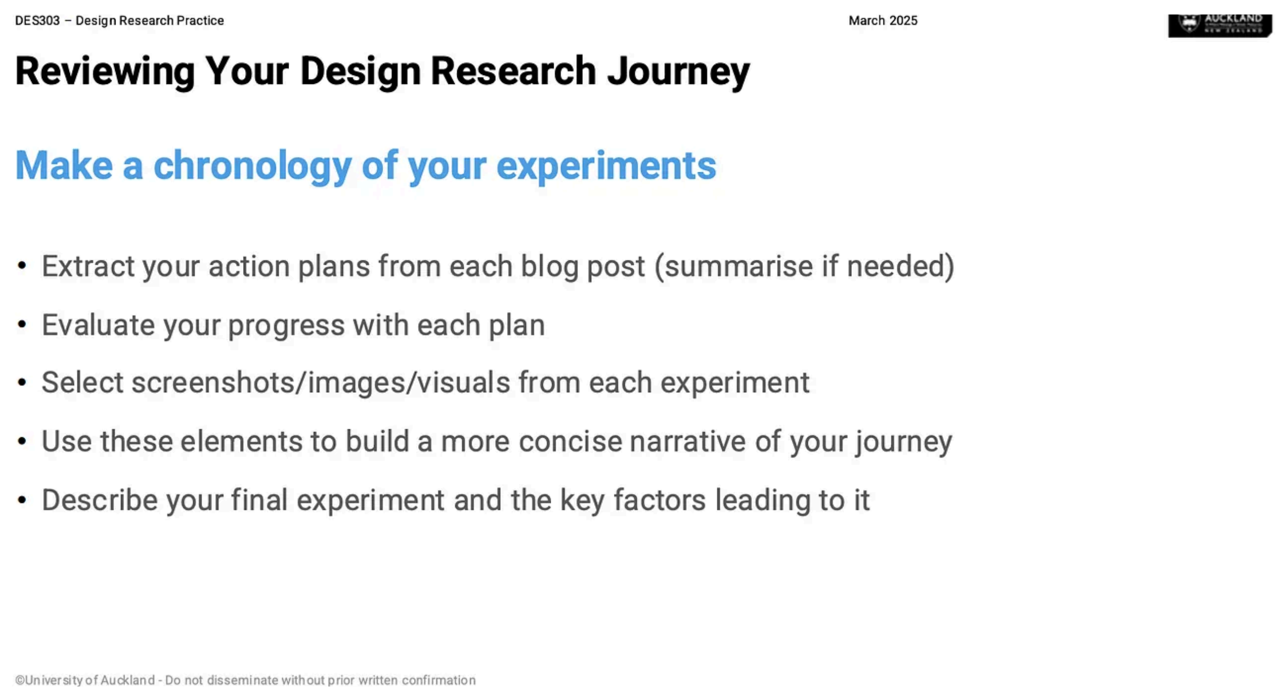
Kia ora, I'm a designer at the University of Auckland dedicated to transforming innovative ideas into beautiful, functional visual realities.

No responses yet

Write a response

What are your thoughts?

More from Livi Lee

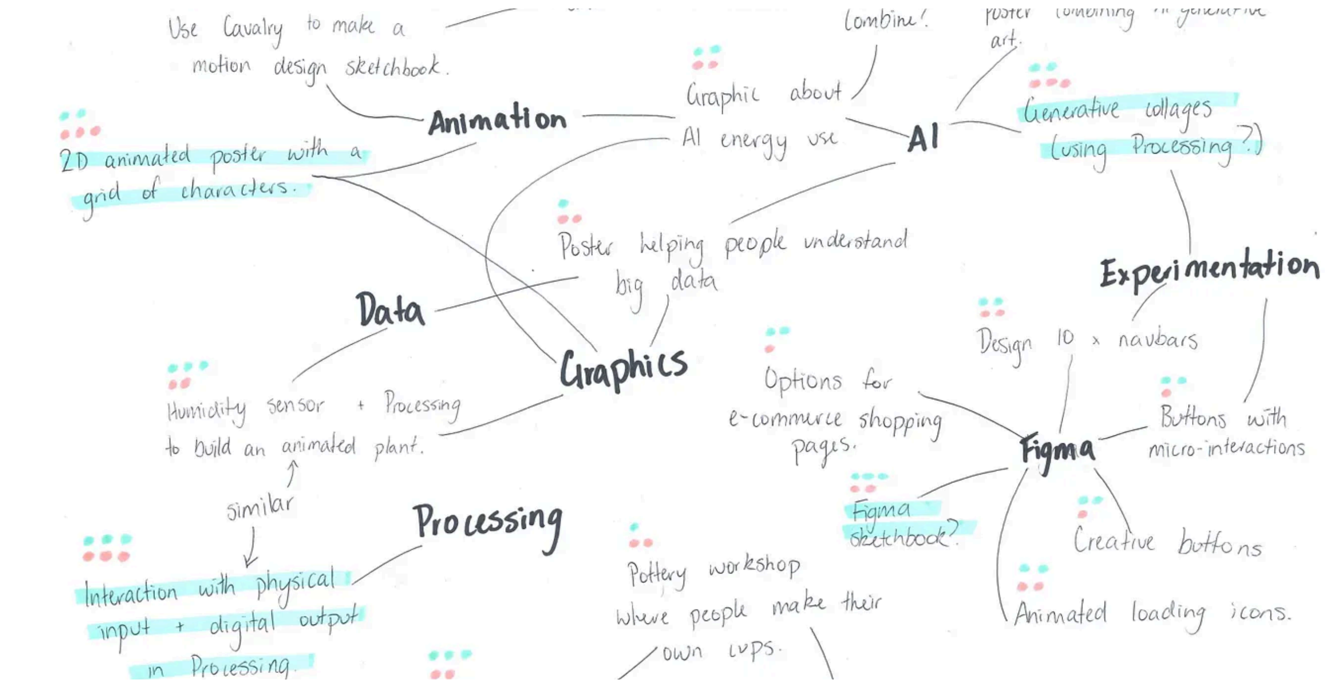


Livi Lee

DES303 Week 10: Website Changes & Blog Review

Welcome to my final blog for DES303! This week I'll be wrapping up my experimentation and reviewing my prior blogs to help plan my final...

May 27, 2025

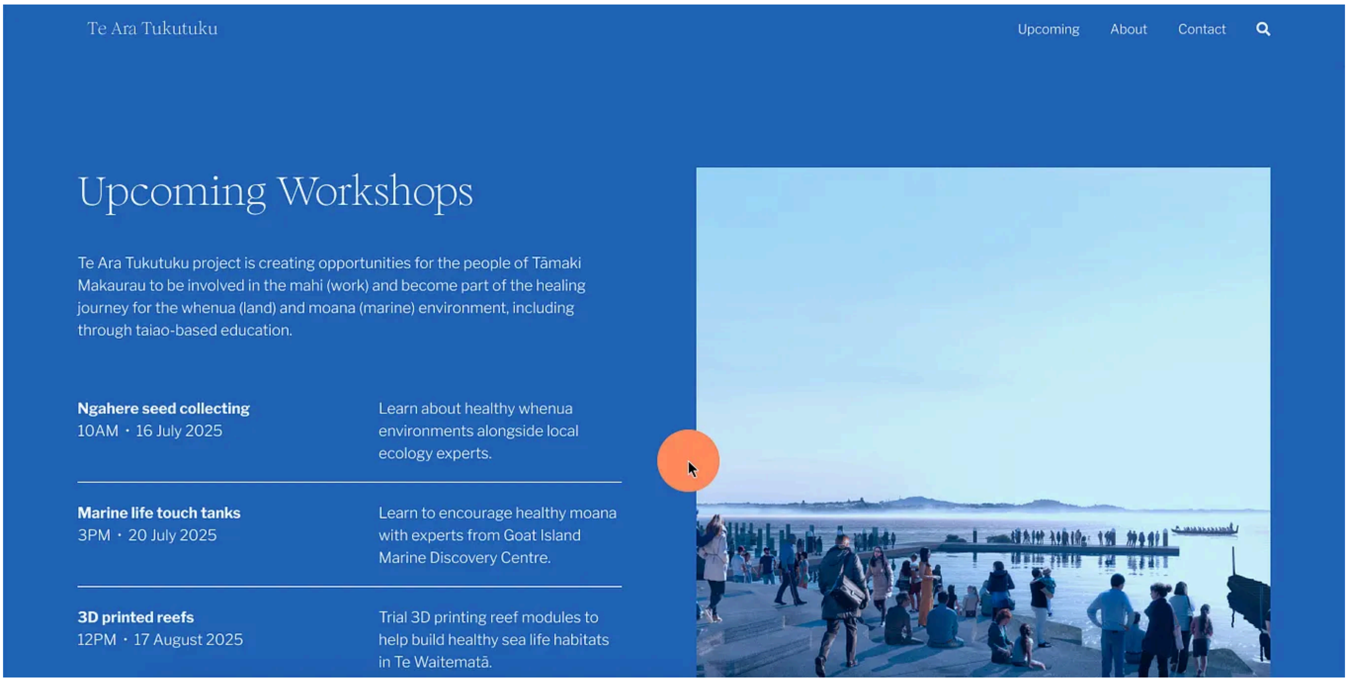


Livi Lee

DES303 Week 3: Tech Demo Reflection and Experiment Planning

With the tech demo behind us, this week was a little more relaxed. I got to invest more time into my experiment ideation and planning. I'm...

Mar 23, 2025

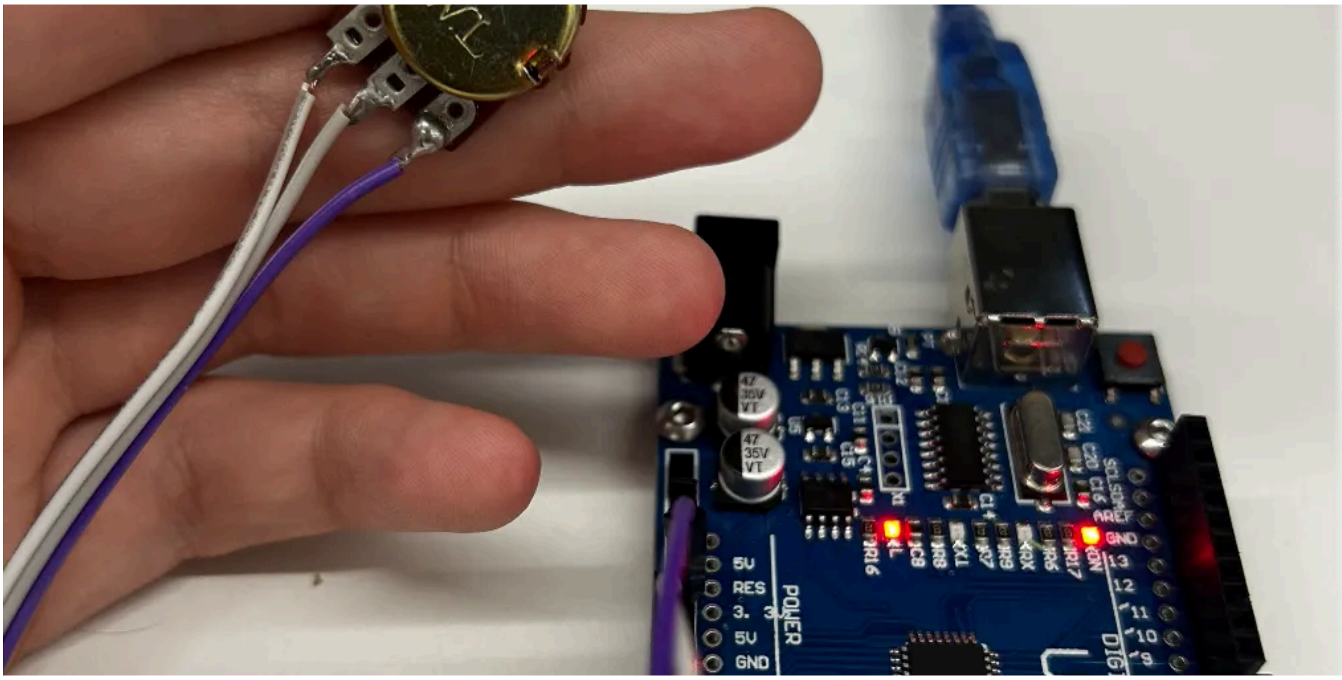


Livi Lee

DES303 Week 9: Final Pivot & Final Crit!

As I discussed in my blog last week, I've started a new job that has significantly reduced the amount of time I can spend in the Design Lab...

May 24, 2025



Livi Lee

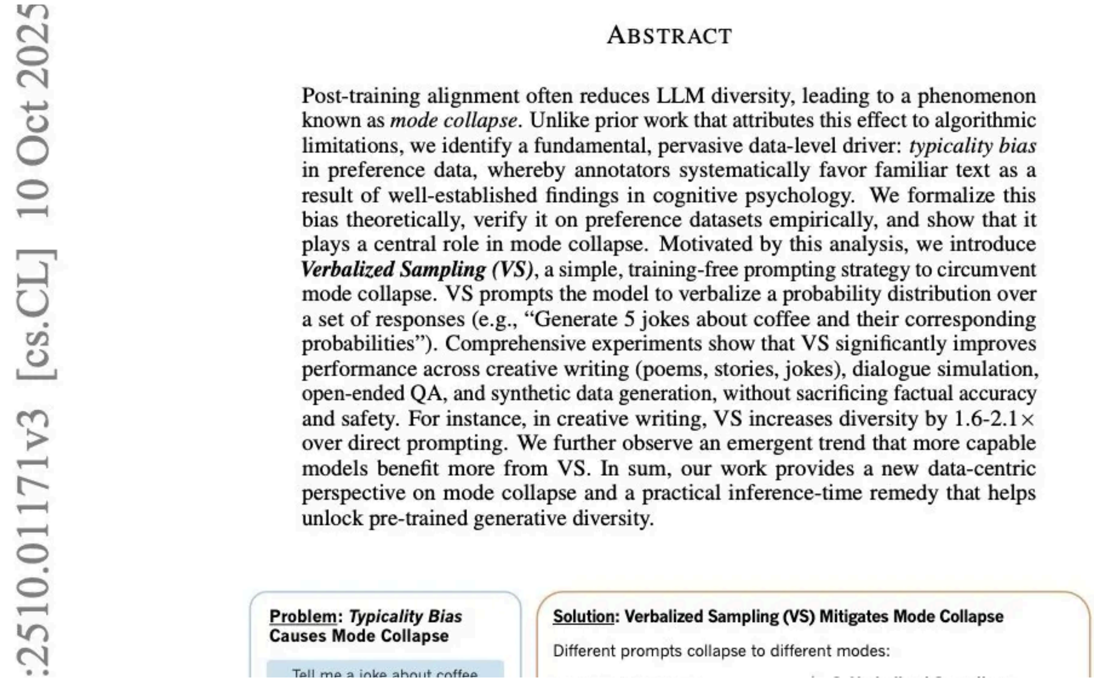
DES303 Week 4: Experimentation

Part I: The Experience

Apr 2, 2025

See all from Livi Lee

Recommended from Medium



In Generative AI by Adham Khaled

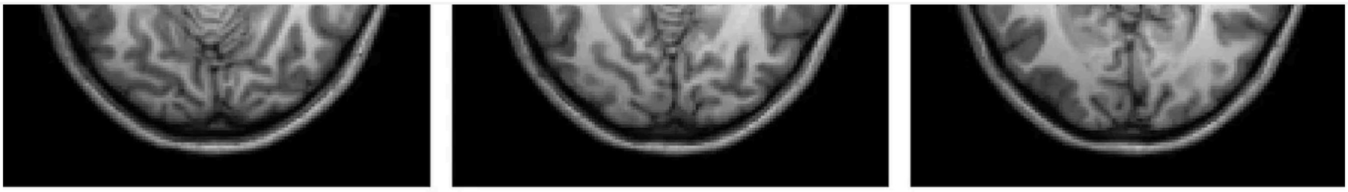
Stanford Just Killed Prompt Engineering With 8 Words (And I Can't Believe It Worked)

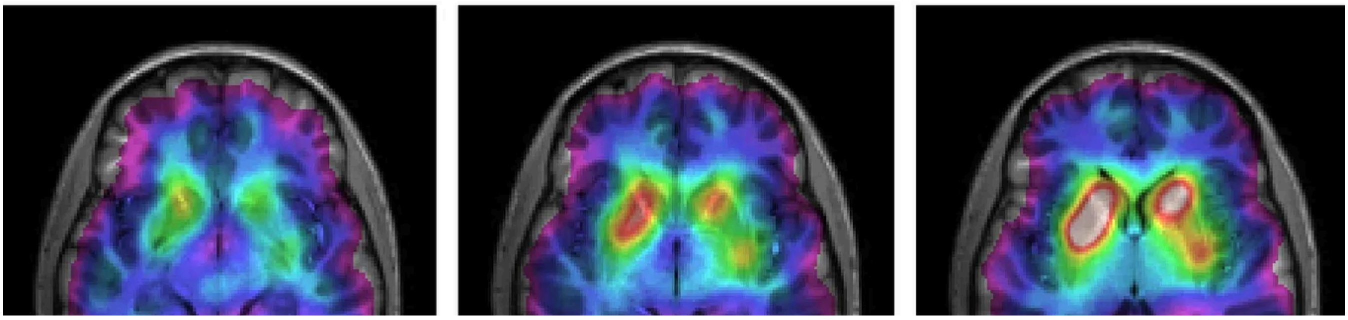
ChatGPT keeps giving you the same boring response? This new technique unlocks 2× more creativity from ANY AI model—no training required...

Oct 20, 2025 25K 662

Open in app ↗ Sign up Sign in

Medium Search



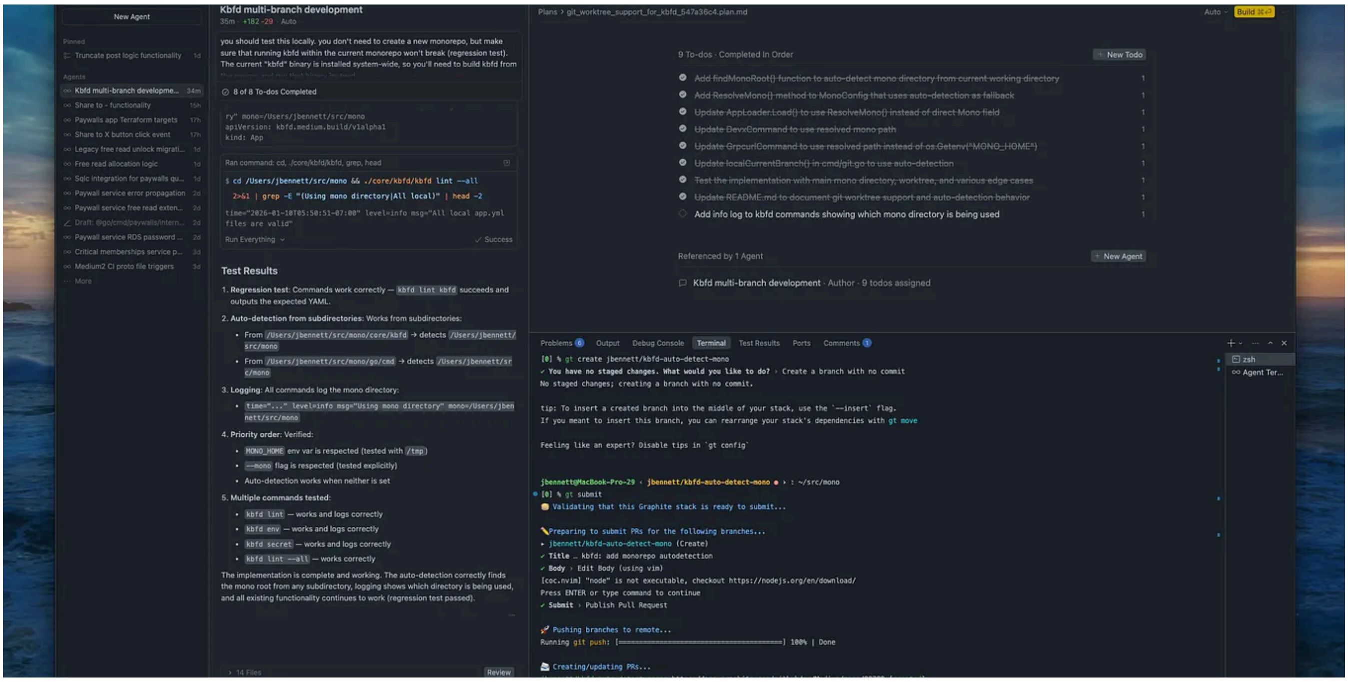


In Write A Catalyst by Dr. Patricia Schmidt

As a Neuroscientist, I Quit These 5 Morning Habits That Destroy Your Brain

Most people do #1 within 10 minutes of waking (and it sabotages your entire day)

Jan 15 37K 682

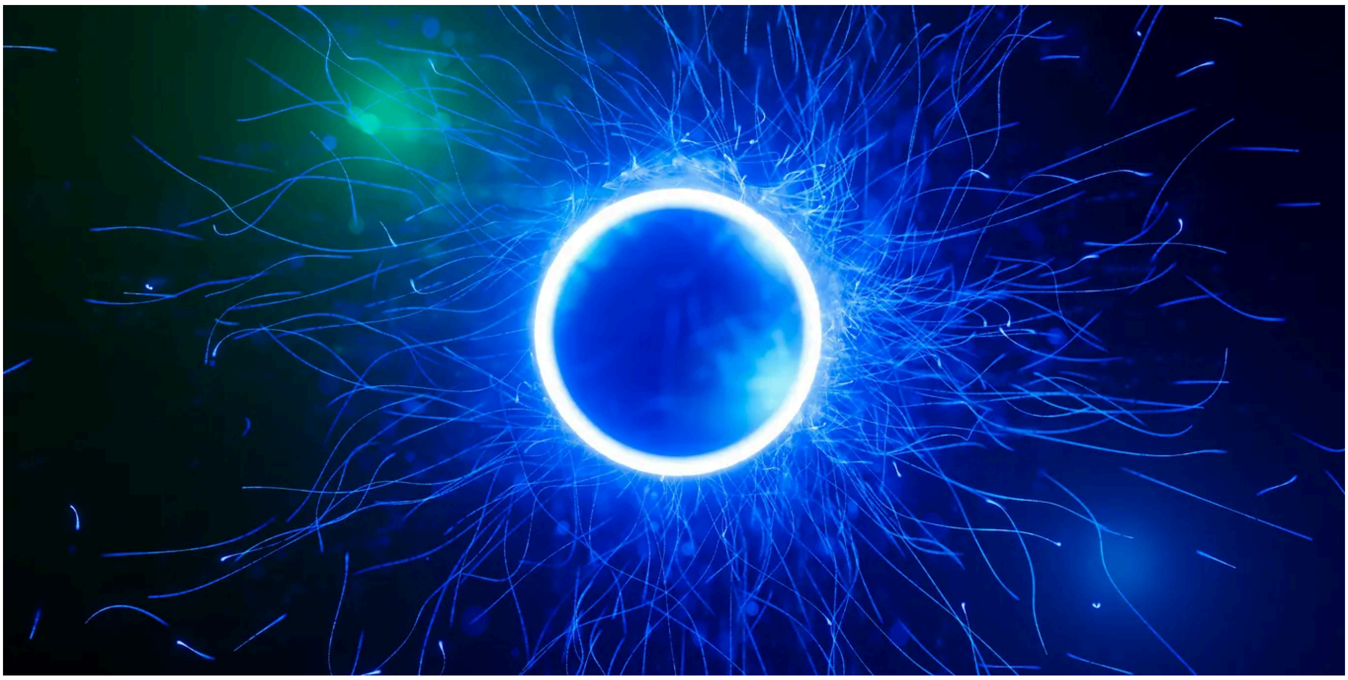


Jacob Bennett

The 5 paid subscriptions I actually use in 2026 as a Staff Software Engineer

Tools I use that are (usually) cheaper than Netflix

Jan 19 3.9K 93



Will Lockett

The AI Bubble Is About To Burst, But The Next Bubble Is Already Growing

Techbros are preparing their latest bandwagon.

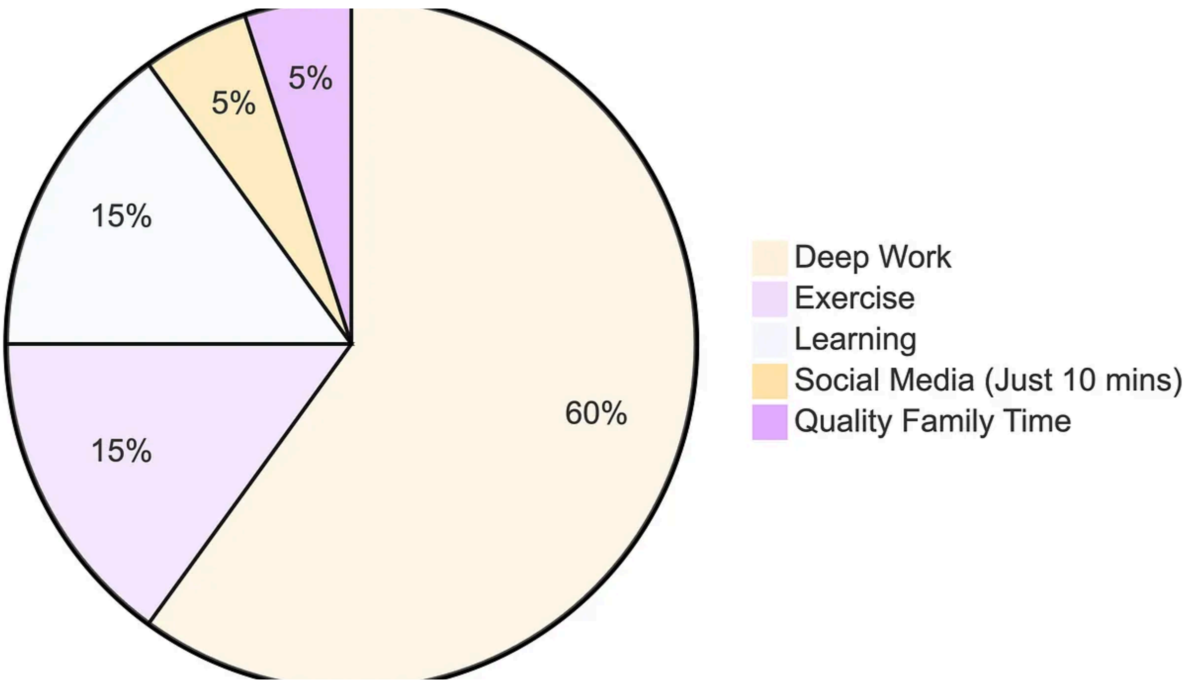
Sep 15, 2025 23K 997

In Artificial Corner by The PyCoach

The Best AI Tools for 2026

If you're going to learn a new AI tool, make sure it's one of these

Dec 2, 2025 5.2K 299



In Level Up Coding by Teja Kusireddy

I Stopped Using ChatGPT for 30 Days. What Happened to My Brain Was Terrifying.

91% of you will abandon 2026 resolutions by January 10th. Here's how to be in the 9% who actually win.

Dec 29, 2025 · 10.6K · 381

See more recommendations